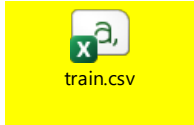


Data Analyst – Internship

Task 3: Titanic Dataset and Exploratory Data Analysis (EDA)

Tool: Python (Pandas, Numpy, Matplotlib and Seaborn)



This exploratory data analysis provides insights into the Titanic dataset including:

- Data types, structure, and missing values
- Distribution of survival outcomes
- Impact of gender and passenger class on survival
- Age and fare distributions
- Correlations between numeric features
- Relationships between key variables and survival status

Step: 1 – Import Libraries & Load Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.pyplot import xlabel
from matplotlib.pyplot import ylabel
from matplotlib.pyplot import show
from matplotlib.pyplot import title
import seaborn as sns
```

✓ 9.2s

Python

```
# Load dataset
df = pd.read_csv("train.csv")

# Basic view
df.head()
```

✓ 0.1s

Python

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Step: 2 – Basic Data Exploration

Step: 2.1 – Dataset Info and Missing Values

Total rows: 891

```
# Dataset Info
df.info()
```

[3] ✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   PassengerId 891 non-null    int64
 1   Survived    891 non-null    int64
 2   Pclass      891 non-null    int64
 3   Name        891 non-null    str
 4   Sex         891 non-null    str
 5   Age        714 non-null    float64
 6   SibSp       891 non-null    int64
 7   Parch       891 non-null    int64
 8   Ticket      891 non-null    str
 9   Fare        891 non-null    float64
10   Cabin       204 non-null    str
11   Embarked    889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

Cabin has large number of missing values

Age has moderate missing values

Embarked has few missing values

```
▶ ✓ # Missing values
    print("Missing Values Count:")
    print(df.isnull().sum())

[15] ✓ 0.0s

... Missing Values Count:
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

Step: 2.2 – Statistical Summary

Average passenger age $\approx$ 29 years

Average fare $\approx$ 32

Age ranges from infants to elderly ($\sim$ 80 years)

Fare has extreme values $\rightarrow$ possible outliers

```
# Statistical Summary
df.describe()
```

✓ 0.0s

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Step: 2.3 – Categorical Distribution

More passengers died than survived

Most passengers were in 3rd class

More male passengers than female

```
▶ # Categorical Distribution
print("Survival Distribution:")
print(df['Survived'].value_counts())
print("\nPassenger Class Distribution:")
print(df['Pclass'].value_counts())
print("\nGender Distribution:")
print(df['Sex'].value_counts())
[5] ✓ 0.0s

... Survival Distribution:
Survived
0    549
1    342
Name: count, dtype: int64

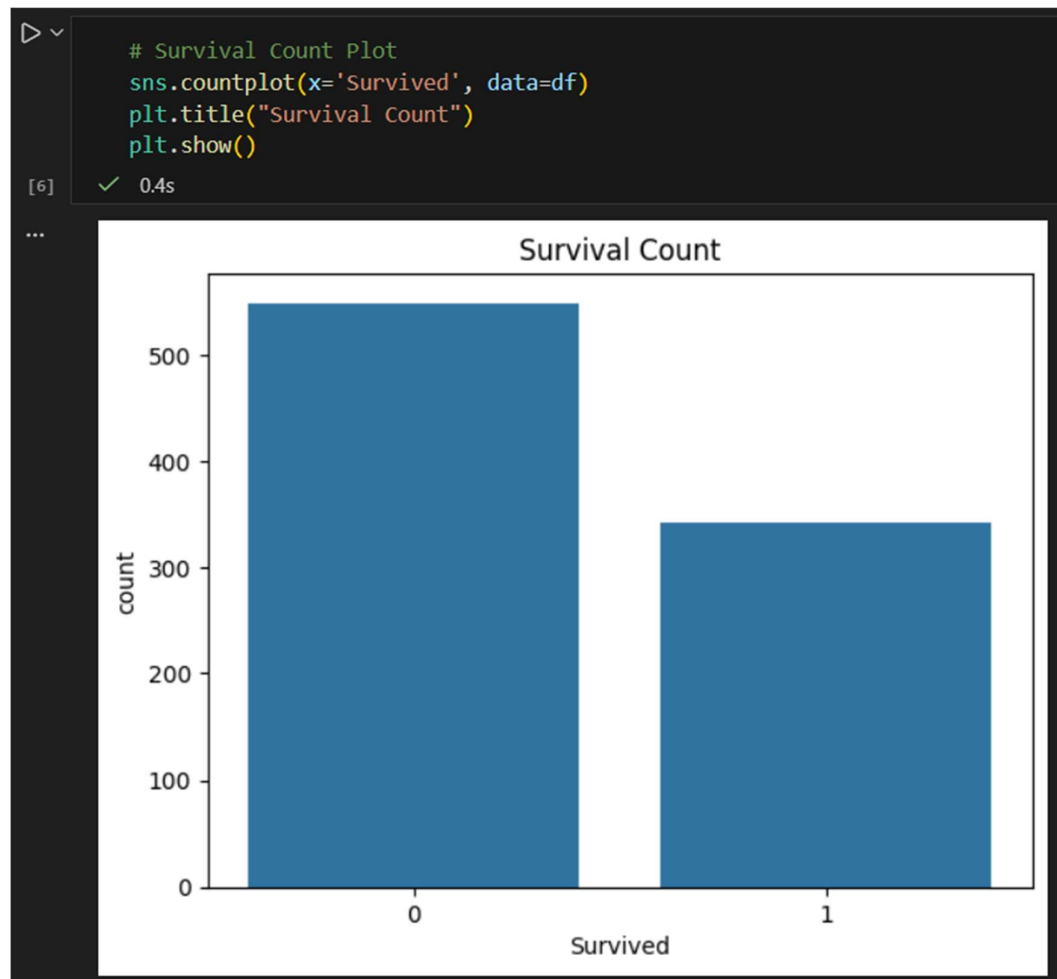
Passenger Class Distribution:
Pclass
3    491
1    216
2    184
Name: count, dtype: int64

Gender Distribution:
Sex
male    577
female  314
Name: count, dtype: int64
```

Step: 3 – Univariate Analysis

Step: 3.1 – Survival Count Plot

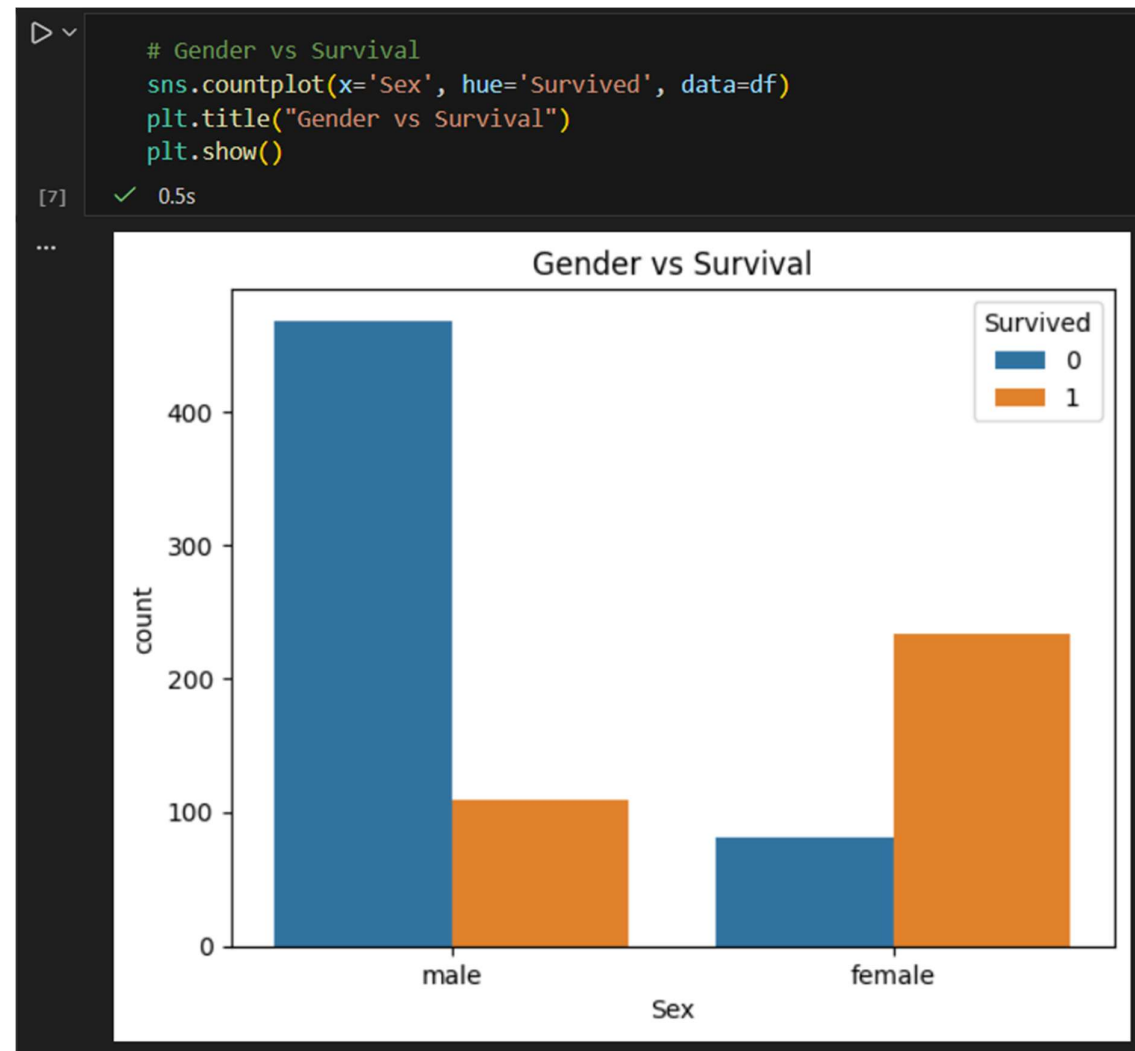
Around 38% survived



Step: 3.2 – Gender vs Survival

Females had significantly higher survival rate

Males had much higher death rate

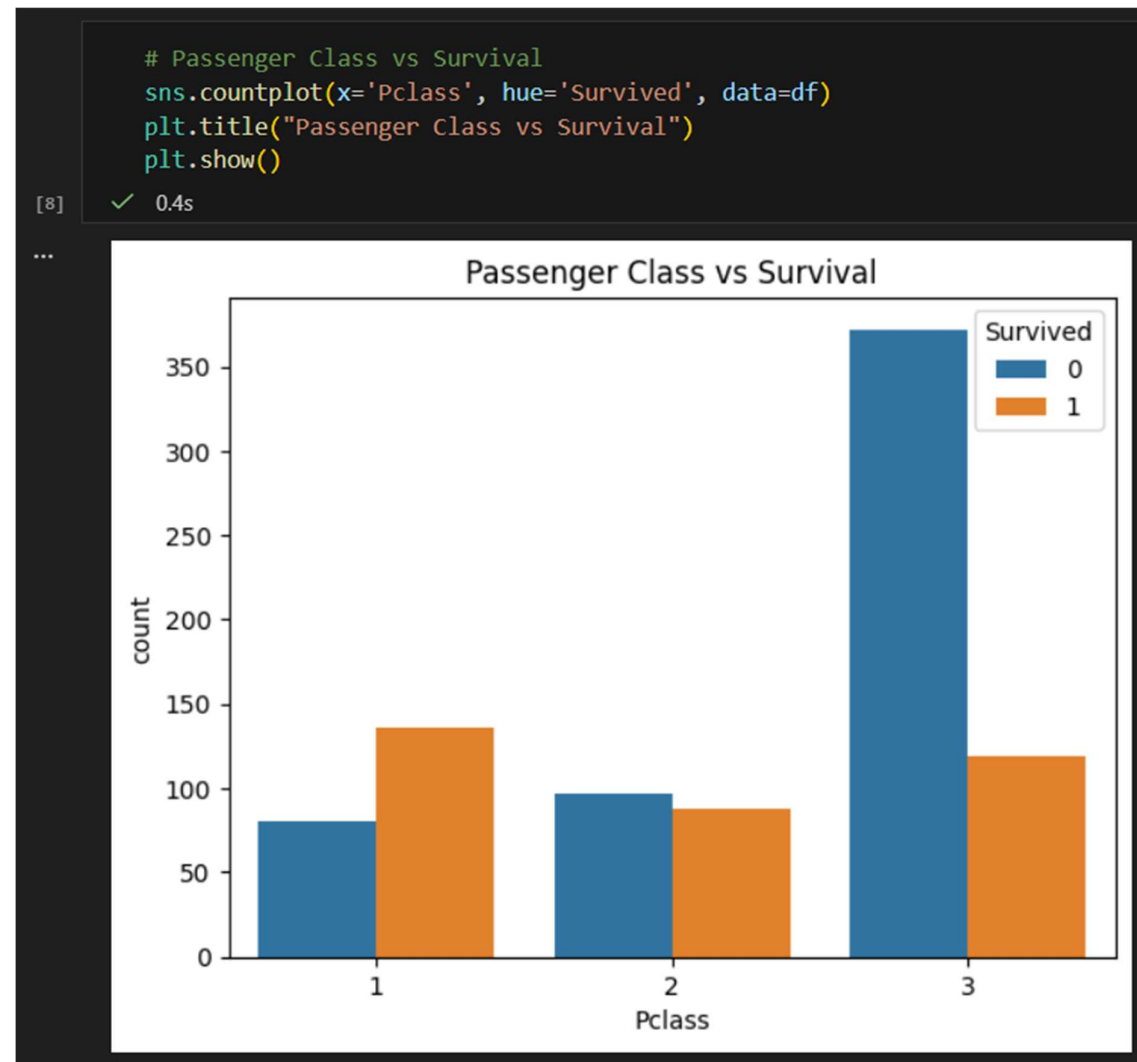


Step: 3.3 – Passenger Class vs Survival

1st class had highest survival rate

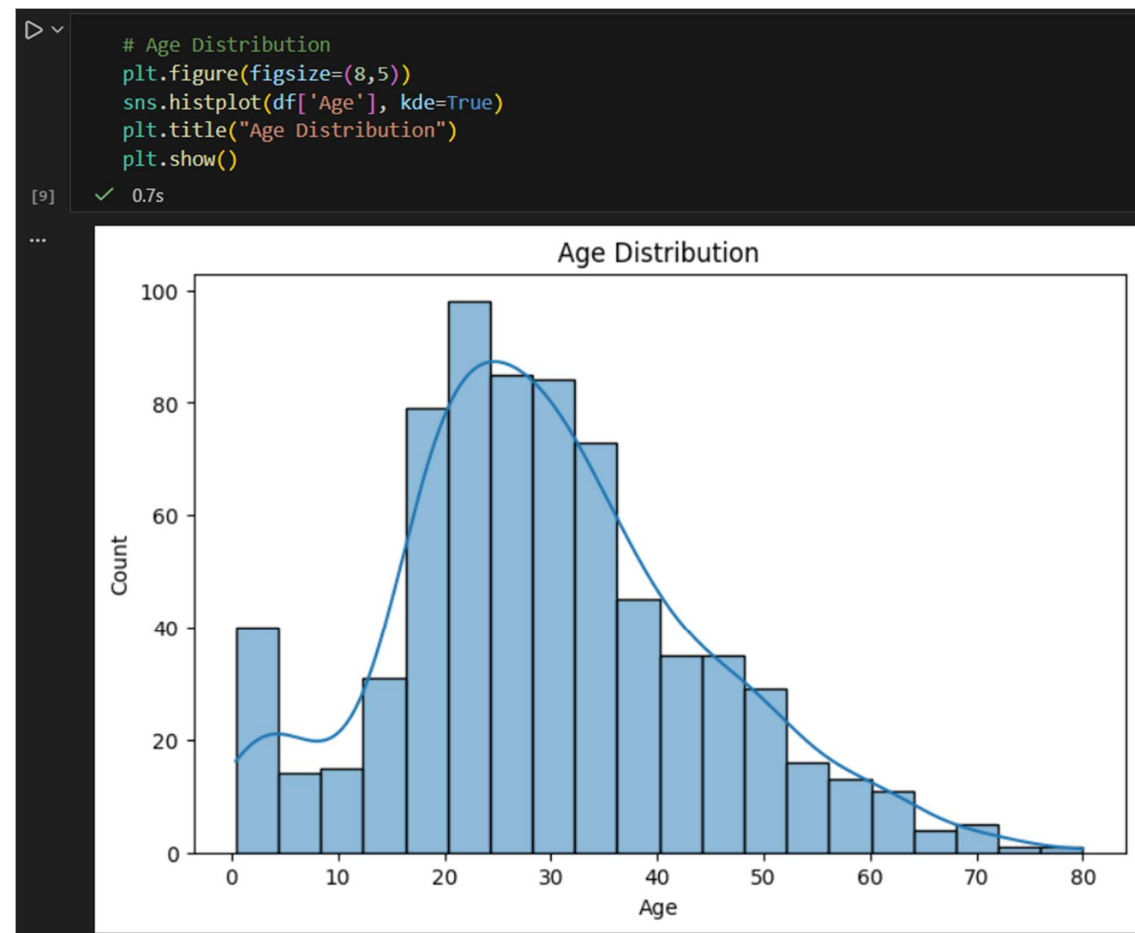
3rd class had highest deaths

Strong relationship between class and survival



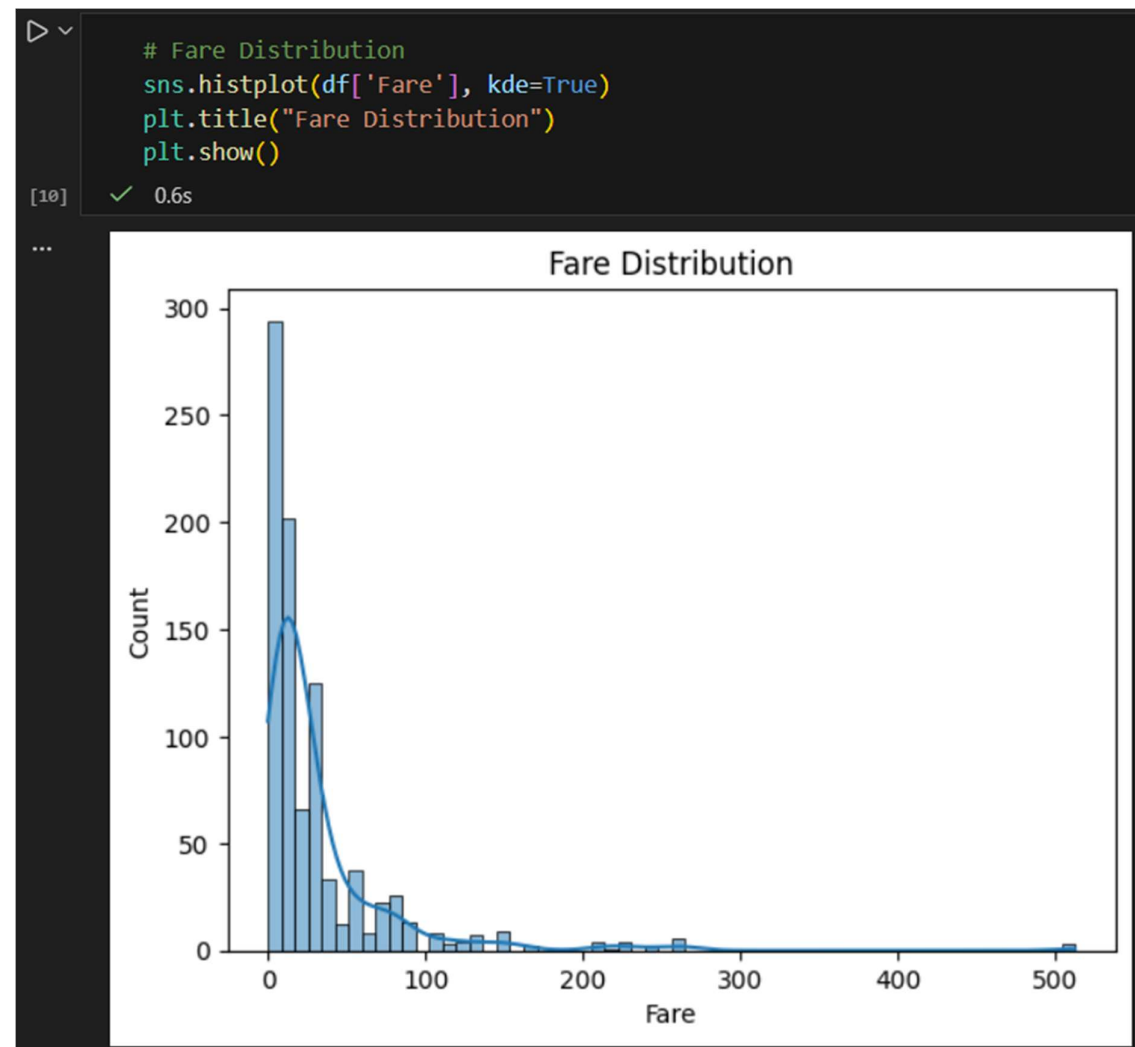
Step: 3.4 – Age Distribution

Most passengers were between 20–35 years



Step: 3.5 – Fare Distribution

Most passengers paid reasonable fares, while a few paid high fares



Step: 4 – Bivariate Analysis

Step: 4.1 – Age vs Survival

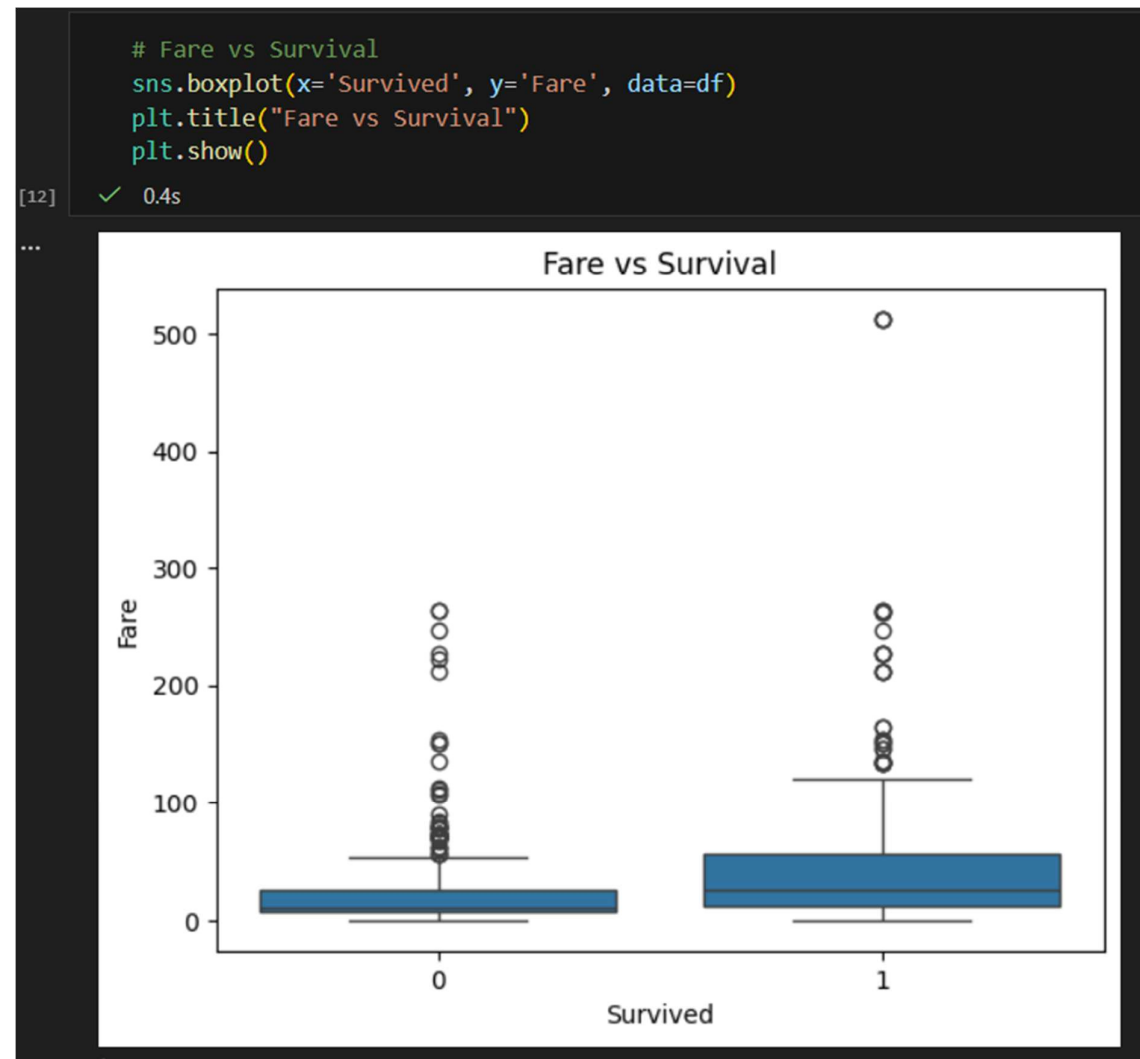
- Children had slightly higher survival rate
- Median age of survivors slightly lower



Step: 4.2 – Fare vs Survival

Survivors paid higher fares on average

Strong correlation between fare and survival



Step: 5 – Correlation Heatmap

Survival positively correlated with:

Fare

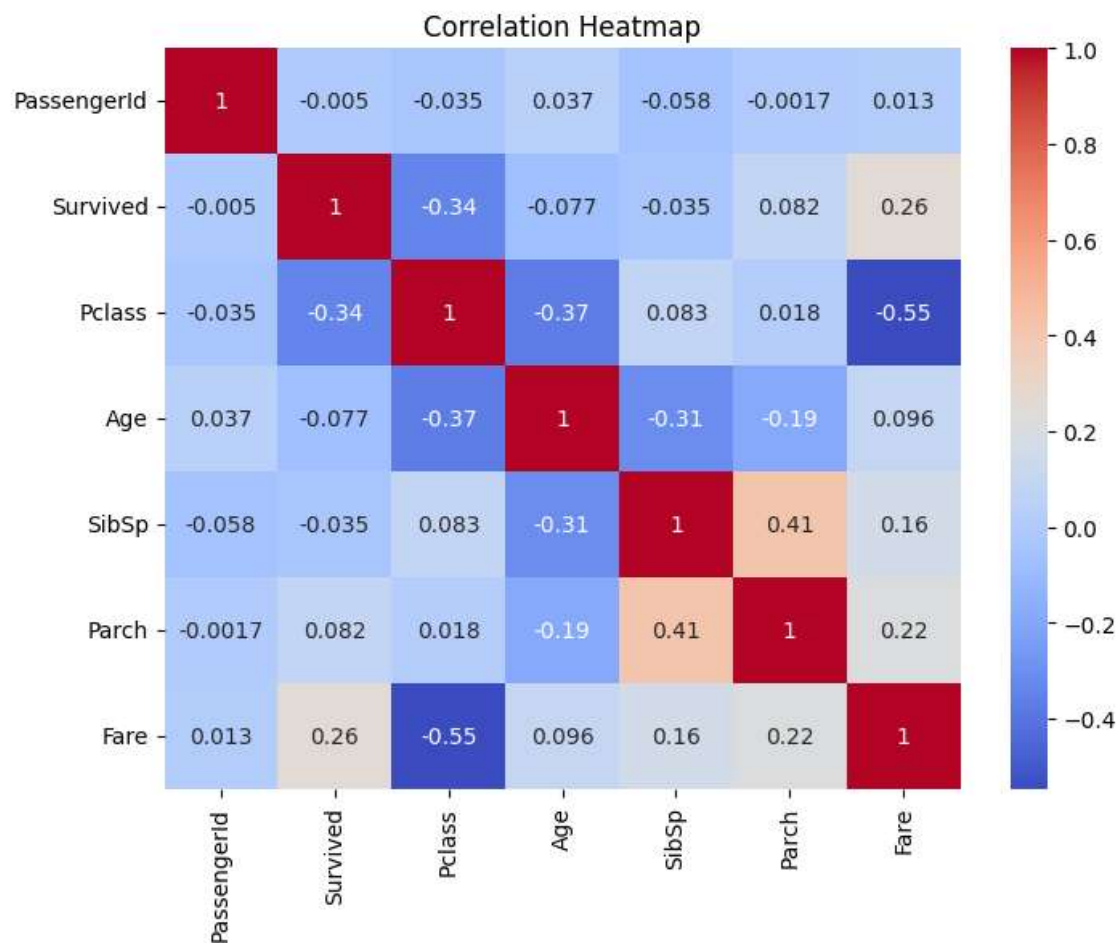
Survival negatively correlated with:

Pclass

Pclass strongly negatively correlated with Fare

```
# Correlation Heatmap
plt.figure(figsize=(8,6))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

[13] ✓ 0.9s



Step: 6 – Pairplot (Relationship Overview)

Clear separation of survival by fare

Higher class passengers cluster in higher fare region

```
# PairPlot Relationship Overview
sns.pairplot(df[['Survived', 'Pclass', 'Age', 'Fare']], hue='Survived')
plt.show()
```

[14] ✓ 4.1s

